

Department Overview

- **Faculty:** Faculty of Life Sciences & Informatics
- **Focus:** The programs are designed to train students in microbiological techniques and provide insight into applied areas of the field, including health sciences, agriculture, environment, marine sciences, and the biotechnology industry.
- **Career Pathways:** Graduates are prepared for careers as scientists in research organizations, clinical laboratory professionals, microbiologists in food processing industries, technologists in biomedical equipment companies, and academicians in public/private universities.
- **Industry Demand:** Microbiologists are in high demand across pharmaceuticals, food and beverage industries, vaccine development, health care products, agriculture, plant/seed prevention, clinical/livestock diagnostics, environmental remediation, and public health laboratories.

1. Bachelor of Science in Microbiology (BS Microbiology)

Admission Requirements

- **Academic Background:** F.Sc (Pre-Medical) with Chemistry and Physics from any recognized board or an equivalent qualification with at least **45% marks**.

Degree Requirements

- **Total Credit Hours:** 134
- **Total Courses:** 42
- **Minimum Academic Standing:** CGPA ≥ 2.0

Curriculum Plan (Program Schema by Semester)

Note: Credit hour structures denote Total Credits (Theory + Lab).

- **1st Semester:** Functional English (3+0), Pakistan Studies (2+0), Mathematics-I (Algebra) (3+0), Biochemistry-I (2+1), Human Physiology-I (2+1), Fundamentals of Microbiology-I (3+1). **Total Credit Hours: 17**
- **2nd Semester:** Communication Skills (3+0), Islamic Studies/Ethics (2+0), Mathematics-II (Calculus) (3+0), Biochemistry-II (2+1), Human Physiology-II (2+1), Fundamentals of Microbiology-II (3+1). **Total Credit Hours: 17**
- **3rd Semester:** Introduction to Computer (2+1), Technical Writing & Presentation Skills (3+0), Genetics (2+1), Ecology and Ecosystem (2+1), Microbial Taxonomy (2+1), Fundamentals of Biotechnology (2+1). **Total Credit Hours: 18**
- **4th Semester:** Biodiversity of Animals (2+1), Current Environmental Issues (3+0), General Virology (2+1), Cell Biology (2+1), Mycology (2+1). **Total Credit Hours: 18**
- **5th Semester:** Biosafety and Risk Management (3+0), Immunology (2+1), Microbial

Anatomy & Physiology (2+1), Epidemiology (2+1), Molecular Biology (2+1). **Total Credit Hours: 15**

- **6th Semester:** Research Methodology (3+0), Soil Microbiology (3+1), Food Microbiology (2+1), Fresh Water Microbiology (2+1), Clinical Bacteriology (3+1). **Total Credit Hours: 18**
- **7th Semester:** Molecular Mechanism of Antimicrobial Drugs (3+1), Marine Microbiology (2+1), Genetics Engineering (3+1), Medical Microbiology (2+1), Applied Immunology (2+1). **Total Credit Hours: 13**
- **8th Semester:** Bacterial Genetics (3+1), Research Project/Internship (0+6), Biostatistics (2+1), Plant Pathology (2+1). **Total Credit Hours: 18**

2. Master of Science in Microbiology (MS Microbiology)

Admission Requirements

- **Academic Background:** BS Microbiology or an equivalent degree with a CGPA $\geq$ 2.0, OR a minimum of 16 years of education in a Life Sciences discipline with at least 50% marks.

Degree Requirements

- **Total Credit Hours:** 30
- **Total Courses Required:** 8
- **Minimum Academic Standing:** CGPA $\geq$ 2.5